

Effect of Onset Position of Ray Casting in Virtual Reality

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ABSTRACT

In virtual reality (VR), interaction techniques significantly influence user experience and performance. In this study, we explore the effect of the onset position of ray casting in VR through a target selection task. We designed the pointing task (i.e., Fitts' law task) for data collection with various onset positions. We then analyzed data to explore how different onset positions affect the individuals' performance for 3D interaction in terms of pointing and selection. Results showed significant effects for each participant yet no generalized impact was found across all participants. Results highlight the complexity of human-computer interaction in ray casting and suggest that a one-size-fits-all approach may not be effective. For future research and practical applications, our findings advocate a personalized approach that takes into account the specific preferences and responses of individual users. This tailored strategy could potentially improve the effectiveness and usability of interactive systems using ray cast techniques.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **HCI theory, concepts and models**.

KEYWORDS

ray-casting, virtual reality, pointing, target selection, Fitts' law

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1 INTRODUCTION

In the evolving landscape of virtual reality (VR), interaction techniques significantly influence user experience and performance [28]. With various 3D interaction metaphors, including object transition, rotation, and scaling [18], the initial and crucial step is the precise identification and designation of the target object, referred to as target selection. However, pointing and selecting distant objects in VR

is still challenging [18, 27]. One of the most prevalent and efficient interaction techniques used to address this issue in VR is ray casting [25]. This method involves virtual hands or VR controllers emitting a ray akin to a real-world laser, allowing users to interact with distant objects such as normal targets, buttons, and menus with high precision [18, 25]. Both virtual hands and VR controllers serve to enrich the user's interaction in VR environments by leveraging ray casting, with each offering unique advantages [22]. Nevertheless, according to previous studies [22], the use of VR controllers is known to have a clear advantage in terms of efficiency in the context of ray casting. Specifically, users experienced reduced physical exertion, a heightened sense of control, and improved speed and accuracy with VR controllers as opposed to virtual hands. Therefore, a lot of studies have been proposed to explore various interaction methods [1, 8, 12, 31] and user modeling [4–6, 20, 24, 35] in VR with ray casting on VR controllers. However, a key yet overlooked parameter in ray casting is the **onset position** of the ray – the point from where the ray originates. This detail is critical because the starting point of the ray can significantly influence its endpoint (see Figure 1) [10, 23], affecting the performance of selecting distant targets. Yet, most existing research either neglects to specify the onset position or lacks rationale for its choice. We argue that careful consideration of the onset position of ray casting could open new opportunities for improving user experience in VR, as different onset points can lead to different interaction dynamics and efficiencies.

VR controllers come in a variety of sizes, including variations in width and height. Specifically, Meta Quest 2 and HTC VIVE Pro controllers have dimensions (unit: cm) of 9×12 and 11×21 , respectively. As depicted in 1, the onset position for ray casting in VR can be identified as a point within the controller space. Hence, the increased area of a controller may lead to large variability in the onset position, which is particularly problematic when accounting for diverse human factors [30], such as the users' grip on the controller and the angle of their hands [17]. In particular, this could impact the user's experience on the alignment of ray and virtual objects, which could be further compounded by the diverse range of motion and dexterity among users [2, 3]. Some might find it easy to direct the ray accurately, while others may struggle with the fine motor control required. This suggests that the inappropriate onset position of ray casting might greatly affect tasks that require high precision, such as selecting targets in VR. The aforementioned factors reveal the difficulty in designing a universally optimal ray casting system and emphasize the importance of understanding the nuances of the onset position and its impact on diverse user groups. Nevertheless, there are no established guidelines for initiating ray casting, which can lead to confusion among practitioners, designers, and researchers, developing and designing VR applications.

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Motivated by this gap, in this paper, we explore the effect of the onset position of ray casting using a commercial VR controller. The purpose of this work is to investigate the substantial impact of onset positions during VR interactions. To this end, we designed the experiment of Fitts' ring similar to the ISO 9241-411:2012 [14] for data collection with various onset positions of ray casting. Afterward, we analyzed the interaction data to explore how different onset positions of ray casting affect the individuals' performance for 3D interaction in terms of pointing and selection.

2 RELATED WORK

Several interaction metaphors could be used for 3D interaction, including mid-air gestures [38], ray casting [10], gaze [36], and sound interaction [19]. However, for touching and manipulating distant objects, ray casting is one of the most effective and efficient interaction techniques [26].

As a fundamental 3D interaction scheme in VR, ray casting has been extensively studied in HCI research. Previous works have mainly focused on interaction techniques for efficiency and user behavior modeling for better UI design in VR, which include gaze-hand alignment for 3D UI selection [31], evaluation of pointing performance with varying ray lengths [4], and exploration of multi-finger ray casting for dense environments [10]. To address interaction challenges with small, distant targets in dense environments, researchers have explored techniques based on ray casting [1, 8, 10, 12, 31]. These include adaptively manipulating rays to support target selection [12], selecting distant targets using controllable cursor [1], employing multiple rays to target large areas [10], and combining gaze with rays to improve precision [8]. In addition, several works have investigated what factors affect the performance of ray casting pointing in VR, including length of ray [4], sound feedback [6], target factor [35] (i.e., Fitts' law [11]), and grip style [5]. However, the onset position of the ray has yet to be examined despite its potential performance impacts. In this paper, we investigate how the onset position affects user performance during ray casting in VR, addressing the gap in existing research.

3 USER STUDY

Through a user study, we aimed to investigate how the ray casting onset position affected users' performance with a target selection task. In our study, onset positions are examined through 3×3 combinations, with variations along the x- and y-axis of the onset point. Throughout the manuscript, we use (x, y) notation to denote a specific onset. Here, the x-axis is partitioned into the leftmost (0), center (50), and rightmost (100), whereas the bottom (0), center (50), and top (100) for the y-axis. Therefore, as shown in Figure 1, the points (0, 0), (50, 50), and (100, 100) correspond to the onset positioned left-bottom, center, and top-right. All experiments were conducted with a Meta Quest 2 head-mounted display (HMD) and right-handheld controller (size: 9 cm width $\times$ 12 cm height). The onset position of the ray is generally determined by the x, y, and z axes in a 3D environment, but in this study, we first investigate the effect of the x and y axes.

3.1 Participants, Design, and Apparatus

3.1.1 Participants. We recruited a total of 10 participants (5 males and 5 females, mean age 25.3(± 1.68)) from a local university. All participants were right-handed and six of them had prior experience with VR applications. They had normal or corrected-to-normal vision. As an incentive for their participation, we provided a coffee voucher.

3.1.2 Experimental design. To see how the onset position of ray casting affects user performance, we implement a pointing and selection task. Our experimental interface was designed as a multi-directional tapping task ISO 9241-411:2012 [14] similar to previous studies [4, 6, 17, 35]. Our Fitts' ring task comprised 15 spherical targets arranged in a circle. We conducted the experiment utilizing a within-subject design structure, comprising three factors: the width of the target (W), the distance to the target (A), and the visual depth (D). The variables had 3 levels each (unit: cm): $W = \{12, 15, 18\}$, $A = \{60, 100, 140\}$, and $D = \{300, 500, 700\}$. This setup resulted in a $3 \times 3 \times 3$ experimental design with all possible combinations of width, distance, and depth for each onset position.

In light of the inherent complexities of a 3D interaction, coupled with factors such as the user's hand tremor, eye fatigue, and arm pain, we opted for relatively wider target widths [15, 16, 21]. In addition, to compensate for the large widths of the targets, which might mitigate the difficulty of our task, we introduced a relatively larger target depth. Through combining W and D , we were able to simulate a variety of complex-sized targets [4, 13, 31, 32, 35]. Each unique combination of W , A , and D was tested in a random order for each onset position (i.e., there were 27 sessions in each onset position). The order of onset positions was also counterbalanced across participants using a balanced Latin Square design [33]. In total, our experiment logs 3,645 selection data (9 onset positions $\times$ 27 sessions $\times$ 15 trials) per each participant. The application for a user study was developed using Unity3D with C#. For the VR experience, we utilized the Meta Quest 2 headset. The experiments were facilitated on a desktop PC equipped with an Intel Core i7-6900k, GTX 1080 Ti, and 16GB RAM.

3.2 Experimental Task and Procedure

3.2.1 Experimental task. In the beginning, the predefined first target was emphasized as blue. When participants selected a blue target, the opposite red sphere turned blue to indicate that it was the next target. No audio or visual feedback was provided before and after selection. After a selection event occurs, the current blue target turns red to show it should not be selected again. The next target on the opposite side then turned blue, signaling the completion of a previous trial and the start of a new selection trial. After 15 repetitions, the Fitts' ring was reshaped with a new combination of target width (W), distance (A), and depth (D). Participants then began a new block of 15 trials starting from the first target.

During the pointing and selection tasks, our application automatically recorded the participants' interaction data - endpoints, movement time (MT), and error - for each trial. We defined an endpoint as a 2D representation (x, y) using vector arithmetic. The centroid of the current target was established as the reference origin, and a vector subtraction was performed with the user's selection point, enabling a simple measurement of spatial displacement along

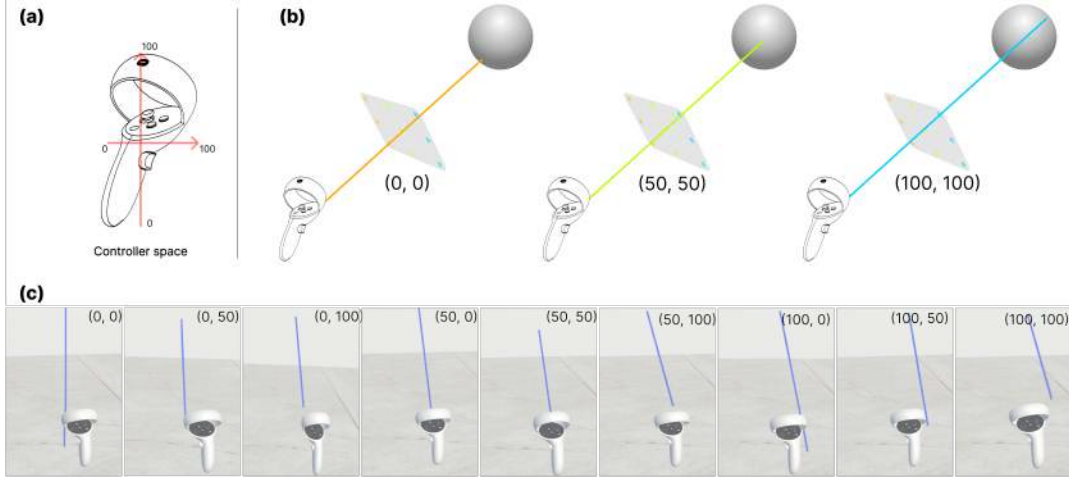


Figure 1: Illustration of various onset positions of ray casting. (a): onset positions are expressed as the ratio of the x- and y-axis in a controller space based on center position (50, 50). (b): different onset position makes a different endpoint during target selection with ray casting. (c): illustration of various onset positions of ray casting in experiment interface.

the x- and y-axes. For MT, we recorded the elapsed time from when the user selected the current point to when they selected the next point regardless of accuracy. Errors were recorded as binary values, indicating the success (1) or failure (0) of each attempt. For each session, we discarded data from the first trial to avoid any potential discrepancy in movement distance (A) according to [17, 32, 35].

In the main experiments, participants used the right handheld controller with visible ray casting extending infinitely. To avoid the "Heisenberg effect" [7, 34] in physical interaction with a controller, target selection is often executed by a non-dominant hand (e.g., pointing with a right-handheld controller using ray casting yet selection with a left-handheld controller). However, in this study, we integrated pointing and selection with only a right-handheld controller for a natural user experience consistent with the existing VR system [35]. Additionally, we did not filter rays to compensate for hand tremors for a collection of natural interaction data within a more realistic environment.

3.2.2 Procedure. Each experiment lasted approximately 60 minutes per participant. First, we introduced the study and experimental tasks without mentioning that the onset position would vary. Participants then provided demographic information and signed a consent form. Prior to the experiments, we conducted practice sessions using a central onset position (i.e., 50% ratio on both the x- and y-axis, as illustrated in the center of Figure 1) and random combinations of W , A , and D to familiarize participants with the environment. For the actual trials, participants completed the selections for one onset position and then took an enforced break. Additional voluntary breaks were allowed as desired. Participants were encouraged to execute the pointing task naturally.

3.3 Measures

To understand the influence of the onset position of ray casting, we employed **effective throughput (eTP)** [29] as a core metric in our study, which measures the efficiency of user interactions

with a computer interface based on both speed and accuracy. **Effective throughput** is calculated as $eTP = ID_e / MT$, where $ID_e = \log_2(A/W_e + 1)$ denotes the effective index of difficulty and MT is movement time. The effective index of difficulty incorporates the error rate by using an effective target width W_e rather than the true physical width W . W_e is calculated as $W_e = SD_{x,y} \times 4.133$, where $SD_{x,y}$ is the standard deviation of participants' selection endpoints in the x and y dimensions. This accounts for accuracy by using the spread of endpoints to determine an effective target width that captures errors. In this way, eTP combines speed (MT) and accuracy (via W_e) to measure overall interaction efficiency. From the single onset position, we obtained 27 eTPs. Each eTP was computed from $eTP = \frac{1}{N} \sum_{i=1}^N eTP_i$, where $N = 14$ represents the number of trials except for first selection within each session. The resulting eTPs ranged from 1.0 and 4.9 bit/s (mean $2.73(\pm 0.64)$). As additional metrics to validate our experiments, as mentioned above, we employed both MT and error rate. MT was used for modeling Fitts' law, and the error rate was used to demonstrate that our experiment was not biased and was not overly easy for participants due to the large width.

4 RESULTS

From data collection, we gathered a total of 34,020 raw data points. Additionally, we removed outliers in terms of both endpoints and MT based on prior works [32, 35, 37] for stable analysis. Using an outlier criterion of three standard deviations from the mean, 100 data points were identified as outliers and subsequently removed, accounting for 0.002% of the data. All data analysis was conducted using Matlab.

4.1 Fitts' law modeling and error rate

We conducted Fitts' law modeling with linear regression for each onset position to confirm whether the user behavior in our experimental setup follows Fitts' law. Our model showed mean $R^2 = 0.91 \pm 0.02$,

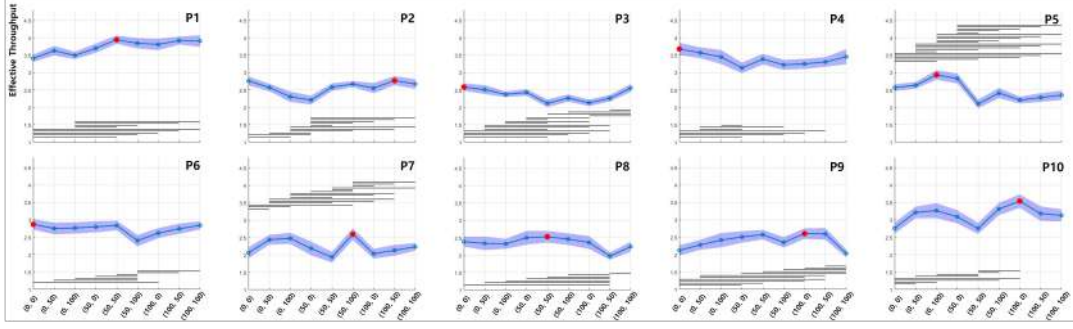


Figure 2: Results of whole participants. The red dot denotes a top eTP among the onset positions and error bands are 95% CIs. Black straight lines denote significant difference ($p < .05$). Each participant may have multiple optimal onset positions if various onset positions are similarly effective and show no significant difference from the top eTP onset.

which reveals a well-fitted and stable result. In addition, participants showed a mean error rate of 0.17 ± 0.09 . Maximum and minimum error rate were 0.34 (P7) and 0.03 (P6), respectively. These results revealed that our experimental setup is non-trivial and can be challenging for specific participants. Additionally, the performance differences among participants indicated that our data reflected individual human factors.

4.2 Influence of different onset position

Before the statistical test, it is crucial to understand our dependent variable (DV) distribution. To evaluate the normality of our DV (i.e., eTPs), we conducted Kolmogorov-Smirnov (KS) tests. The results indicated that the data in all conditions followed a normal distribution.

4.2.1 Within-participant results. Figure 2 illustrates the results of the whole participants. Based on the normality test results, we fitted linear mixed-effect models (LME) with eTPs for each participant to determine the effect of onset position on performance. As parameters for LME, onset position was fixed effects, (50, 50) as a reference (i.e., baseline), since the onset at the center position is commonly used in the current VR applications [12, 22]. Factor combinations provided in random order were random effects. To validate the LME, we employed a Type II Wald χ^2 test. For post hoc comparisons, we used Bonferroni-adjusted pairwise tests to confirm significant differences between all onset positions. As a result, Type II Wald χ^2 test with LME revealed significant differences in onset positions in all participants ($p < .001$). This means that all participants have performance differences among different onset positions compared to a baseline position. Furthermore, as indicated in Figure 2, our post hoc comparisons showed multiple significant differences among different onset positions within each participant. Also, it should be noted the best onset position yielding the top eTP depends on the participant. However, in contrast to our findings, previous research, including [12], typically used the central position (i.e., (50, 50)) as their default onset. Moreover, this approach was commonly used in major game development platforms like Unity (i.e., anchor transform). The results presented in this section, however, suggest that finding the optimal onset position for an individual should be considered to improve performance and user experience.

On the other hand, our results were collected from a limited number of trials and, therefore, may be susceptible to learning effects as pointed out by [17]. For a more in-depth understanding of the impact of onset position on performance, we determined the order effect using a two-way repeated measures analysis of variance (RM-ANOVA) grouped by a condition order and onset positions for each participant. The observations were highly dependent on each individual, so we chose to focus on the effect size for each factor ($\eta_p^2 > 0.14$ indicates large significance [9])). Basically, if the onset position does not hold individual significance, all statistical significance should stem from learning effects (i.e., condition order). As a result, we could observe that every participant showed large effect sizes ($p < .001$, $\eta_p^2 > 0.14$) in terms of onset position, condition order, and their interaction effect. Even though there was an effect for the order of the conditions, we could also find a significant effect for the onset position. Particularly, three participants (P5, P7, and P8) revealed larger effect sizes for the onset position (i.e., 0.51, 0.38, and 0.33, respectively) than the condition order (i.e., 0.5, 0.24, and 0.3, respectively). For certain individuals, the onset position plays a more crucial role than the order of condition order for task performance. However, for a comprehensive understanding of the impact of onset positions on user behavior, further research with an increased sample size is imperative.

4.2.2 Whole-participant results. Beyond the within-participant results, we proceeded with RM-ANOVA on averaged eTP to find out any generalizable pattern over all participants. The result of Mauchly's test of sphericity for eTPs indicated no violation ($\chi^2(35) = 44.359$, $p > 0.05$). In contrast to within-participant analysis, the results of RM-ANOVA found no significant differences between onset positions ($F_{(8,72)} = 0.28$, $p > 0.05$). This implies that variations in performances among different participants led to a cancellation of the effects associated with different onset positions, making it challenging to identify any distinct pattern. In particular, the absence of a general influence of onset position poses a problem for practitioners and researchers using ray casting. This would result in the lack of universal guidelines on the onset position for VR applications, which makes it more challenging to design a ray casting interaction that satisfies all users.

Table 1: Result of two-way RM-ANOVA. $X \times Y$ denotes the interaction effect between two factors. For the η_p^2 , an effect size larger than 0.14 signifies the variable has a large effect [9].

Participant	IV	F	p	η_p^2	IV	F	p	η_p^2	IV	F	p	η_p^2
1	X	26.79	.000	0.50	Y	8.66	.000	0.24	$X \times Y$	0.40	.806	0.01
2	X	5.53	.006	0.17	Y	4.58	.014	0.15	$X \times Y$	20.79	.000	0.44
3	X	21.17	.000	0.44	Y	3.09	.053	0.10	$X \times Y$	22.94	.000	0.46
4	X	15.29	.000	0.37	Y	0.59	.552	0.02	$X \times Y$	3.47	.010	0.11
5	X	49.43	.000	0.65	Y	15.80	.000	0.37	$X \times Y$	22.78	.000	0.46
6	X	3.02	.056	0.10	Y	5.19	.008	0.16	$X \times Y$	11.51	.000	0.30
7	X	8.55	.000	0.24	Y	41.84	.000	0.61	$X \times Y$	11.49	.000	0.30
8	X	14.75	.000	0.36	Y	2.59	.085	0.09	$X \times Y$	5.22	.000	0.16
9	X	10.05	.000	0.27	Y	12.98	.000	0.33	$X \times Y$	14.51	.000	0.35
10	X	7.39	.001	0.22	Y	4.64	.013	0.15	$X \times Y$	10.05	.000	0.27

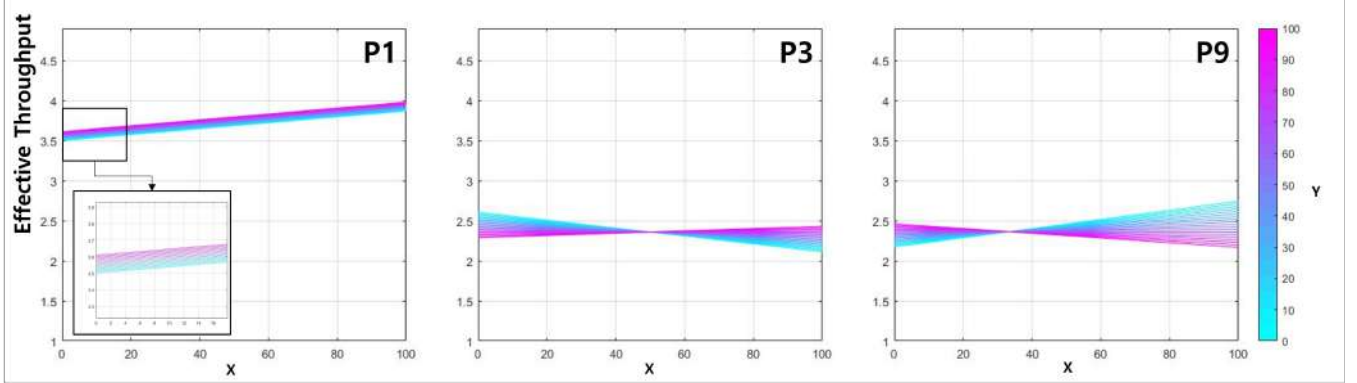


Figure 3: Simulation of effective throughput on onset position.

4.2.3 Which axis more affected the performance? With the aforementioned experimental results, it becomes evident that characterizing the impact of onset position is an intricate and challenging matter, posing difficulties in generalization. For a more in-depth analysis, we attempted to determine which aspects of the onset position affected each participant. For this, we conducted a two-way RM-ANOVA for each individual, using the x- and y-axis in the onset position as an independent variable (IV; X and Y). No violations were found in Mauchly's sphericity tests for each participant. The comprehensive results of the two-way RM-ANOVA are presented in Table 1. Note that this result only indicates the internal variability of each subject and does not allow for generalization. Given the dependence of the data on individual participants, our analysis was particularly attentive to the effect size as an alternative to the p-value. The findings indicate that the factors influencing outcomes vary uniquely across participants, highlighting the importance of considering participant-specific effects. Specifically, most participants except P4 ($F_{(2,52)} = 15.29, p < .001$ in X-axis only) were affected by more than two factors among X, Y, and their interaction with large effect size ($\eta_p^2 > 0.14$). These results represent the complexity inherent in human-computer interaction within VR ray casting interfaces, implying that a universal, one-size-fits-all approach may be ineffective. Therefore, a tailored approach to design factors should be required to optimize the user experience of VR interaction based on ray casting.

5 PERFORMANCE SIMULATION ON DIFFERENT ONSET POSITIONS

In this section, we modeled ray casting behavior by varying onset positions from 0% to 100% with a 5% step in both the X- and Y-axis to investigate the impact of onset position on eTP. To this end, we fitted simple linear regression (LR) to predict eTP on different onset positions as well as further analyzed user behavior. The eTPs were modeled using significant IV for each participant based on the results of two-way RM-ANOVA (Table 1). A LR model was used for each participant as follows: $eTP = a_0 + a_1X + a_2Y$ for P1, $eTP = a_0 + a_1X + a_2(X \times Y)$ for P3 and P8, $eTP = a_0 + a_1X$ for P4, $eTP = a_0 + a_1Y + a_2(X \times Y)$ for P6, and $eTP = a_0 + a_1X + a_2Y + a_3(X \times Y)$ for the remaining participants. Each coefficient a_0 to a_3 is independent across all participants, resulting in 10 linear regression models being fitted. All intercepts a_0 were set as eTP of the baseline (i.e., central) onset position of each subject.

As a result, LR models did not demonstrate a strong fit across all participants, most of them with adjusted R^2 values < 0.7 , indicating the model's limited ability to explain data variability. Despite the statistical significance and a sufficiently large effect size observed in our data, it became evident that accurately modeling performance based on the onset position is very complex and challenging. Among the participants, P1, P3, and P9 exhibited relatively better model fit compared to others, with adjusted R^2 values of 0.64, 0.45, and

0.53, respectively, despite the presence of complex data patterns from onset positions. Therefore, we conducted further analysis of simulation data for P1, P3, and P9. Figure 3 illustrates the simulation results. For these participants, we obtained consistent results with the findings described in Section 4.2.1. Even in the simulation data, the observed optimal positions in terms of eTP corresponded with those found in experimental data.

The following are possible guidelines based on the simulation data for determining onset positions for these participants. First, P1 was affected by X and Y, but there was no interaction effect. For P1, we found $eTP = 3.74 + 0.04X + 0.01Y$ from model fit. The simulation model from P1 revealed that onset position and eTP have a direct proportional relationship. Therefore, it is recommended to set up-right onset position for P1. Compared to P1, P3 and P9 exhibited more complex patterns, where the relationship between onset position and eTP changes based on the position of onset. For P3 and P9, we found $eTP = 2.35 - 0.01X + 0.007(X \times Y)$ and $eTP = 2.38 + 0.01X - 0.01Y + 0.007(X \times Y)$ from the model fit, respectively. Figure 3 illustrates that the impact of onset position on eTP shows a conflicting pattern depending on X. To avoid any unintended negative consequences for eTP, attempts to increase the positive factors for P3 and P9 should be carefully made with a consideration of the interaction effects as well as a balance with the potential increase in negatively affecting factors.

Although the performances from various onset positions are complex and inconsistent, we could observe some participants' data were well-fitted using simple linear regression. This simulation shows a potential approach to personalization and customization of onset position for enhancing user experience and performance during 3D interaction based on ray casting. With future improvement, we believe that prediction of the user-specific performance of ray casting interaction would be possible.

6 CONCLUSION AND FUTURE WORK

In this study, we explored the effect of the onset position of ray casting in VR using a target selection task. Results showed that the onset position revealed significant effects for each participant yet no generalized impact was found across all participants. In addition, all participants except one showed sensitivity to multiple factors, including X, Y, and their interaction. These findings present new challenges for VR practitioners and researchers to consider the human or environmental factors when utilizing ray casting. Our results advocate for personalized, tailored approaches accounting for users' specific preferences and responses when employing ray casting in practical applications or future research. Rather than a one-size-fits-all strategy, this idiographic perspective could enhance the usability and efficacy of interactive ray casting systems by optimizing them to individuals' needs. Further studies on larger and more diverse samples are needed to deepen our understanding of how onset position can be leveraged to improve user experiences.

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